

Boyuan Liang

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EDUCATION

University of California, Berkeley

Aug 2022 – Present

Ph.D. Candidate in Mechanical Engineering

Berkeley, CA

- Advised by Prof Masayoshi Tomizuka
- Major in Controls, Minor in Design and Applied Mathematics

National University of Singapore

Aug. 2017 – June 2021

B.Sc. (Hons.) with Highest Distinction in Applied Mathematics

Singapore

- Medal of Excellence in Mathematics (Top 1% of Batch 2021)

EXPERIENCE

Autel US

May 2026 – August 2026

Research Intern, Robotics

Cambridge, MA

- Developed a piezoelectric haptic display with more compact pixels and driver systems than the SOTA. Conducted volunteer tests for design verifications.
- Built a haptic teleoperation system for long-horizon contact-rich task data collection.

Mitsubishi Electric Research Laboratories

May 2023 – August 2023

Research Intern, Robotics

Cambridge, MA

- Developed an uncertainty-aware robust planning framework for contact-rich manipulation, utilizing second-order limit surface models to maintain stable environmental contact.
- Validated the algorithm on a 6-DOF industrial arm on in-hand manipulation tasks under physical parameter errors, achieved 100% success rate.

A*STAR Institute for Infocomm Research

July 2021 – July 2022

Research Engineer, Robotics

Singapore

- Developed a parameterized particle filtering framework for online tactile in-hand pose and shape estimation, achieved sub-millimeter/degree accuracy.
- Designed a tactile-guided dynamic manipulation framework using gain-scheduling PD control and a GRU-based deep learning model to precisely hit unknown objects to target distances with less than 10% error.

RECENT PROJECTS

Piezoelectric Haptic Display | PCB Design, CAD, CAM

On-going

- Designed a haptic display using piezoelectric vibrators as tactile pixels for augmented reality and robotic teleoperation applications.
- Engineered a driver PCB and integrated ESP32 for a high-density piezoelectric array. Implemented high-speed SPI communication and optimized thermal management for compact integration.
- Executed comprehensive signal integrity and timing analysis using oscilloscopes and logic analyzers to achieve deterministic sub-100 μ s control latency for pixel vibration frequency.
- Conducted volunteer experiments to verify effectiveness of the design.
- *On-going project. Details are currently under pending publication.*

TECHNICAL SKILLS

Electronics: PCB Design (Altium Designer/KiCAD), MCU Coding (ESP32/Arduino), Circuit Analysis (LTSpice/TINA-TI)

Mechanical Design: CAD and CAM (Solidworks/Fusion 360), 3D CNC, Waterjet, Laser Cutting

Programming: C/C++, MATLAB/Simulink, Python, LabVIEW

PUBLICATIONS

- J. Zhou, **B. Liang**, J. Huang, H. Zhang, I. Zhang, Y. Wang, P. Abbeel, M. Tomizuka, "Singularity-Avoidant Motion Mappings for Pose-Tracking Teleoperation with Operational Jacobian Analysis", *IEEE/ASME Transactions on Mechatronics*, Early Access, 2026
- J. Zhou, W. Chen, J. Huang, **B. Liang**, Y. Liu, M. Tomizuka, "Programmable Locking Cells (PLC) for Modular Robots with High Stiffness Tunability and Morphological Adaptability", *IEEE Transaction on Robotics*, vol. 41, pp. 6460-6477, 2025
- J. Zhou, J. Huang, **B. Liang**, X. Zhang, X. Ma, M. Tomizuka, "Prismatic-Bending Transformable (PBT) Joint for a Modular, Foldable Manipulator with Enhanced Reachability and Dexterity", *IEEE/ASME Transaction on Mechatronics*, vol. 30, no. 4 and in *International Conference on Advanced Intelligent Mechatronics*, 2025
- J. Zhou, **B. Liang**, J. Huang, M. Tomizuka, "Unified Manipulability and Compliance Analysis of Modular Soft-Rigid Hybrid Fingers", in *Modeling, Estimation and Control Conference*, 2025
- T. Zhang, Z. Wu, Y. Chen, Y. Wang, **B. Liang**, S. Moura, M. Tomizuka, M. Ding, W. Zhan "Physics-Aware Robotic Palletization with Online Masking Inference", in *IEEE International Conference on Robotics and Automation*, 2025
- **B. Liang**, L. Sun, X. Zhu, B. Zhang, K. Sreenath, M. Tomizuka, "Adaptive Energy Regularization for Autonomous Gait Transition and Energy-Efficient Quadruped Locomotion", in *IEEE International Conference on Robotics and Automation*, 2025
- M. Yu, **B. Liang**, X. Zhang, X. Zhu, X. Li, M. Tomizuka, "In-Hand Following of Deformable Linear Objects Using Dexterous Fingers with Tactile Sensing", in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024
- **B. Liang**, K. Ota, M. Tomizuka, and D. K. Jha, "Robust In-Hand Manipulation with Extrinsic Contacts", in *IEEE International Conference on Robotics and Automation*, 2024
- H. Wang, W. Liang, **B. Liang**, H. Ren, Z. Du, and Y. Wu, "Robust Position Control of a Continuum Manipulator Based on Selective Approach and Koopman Operator", *IEEE Transactions on Industrial Electronics*, vol. 70, no. 12, pp. 12522-12532, 2023.
- **B. Liang**, W. Liang, and Y. Wu, "Tactile-Guided Dynamic Object Planar Manipulation", in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.
- **B. Liang**, W. Liang, and Y. Wu, "Parameterized Particle Filtering for Tactile-Based Simultaneous Pose and Shape Estimation", *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 1270-1277, 2022.

AWARDS AND SCHOLARSHIPS

National Science Scholarship (Ph.D.)
*Agency of Science, Technology and Research (A*STAR)*

August 2022 – July 2027
Singapore

NUS Science and Technology Scholarship
National University of Singapore

July 2017 – June 2021
Singapore